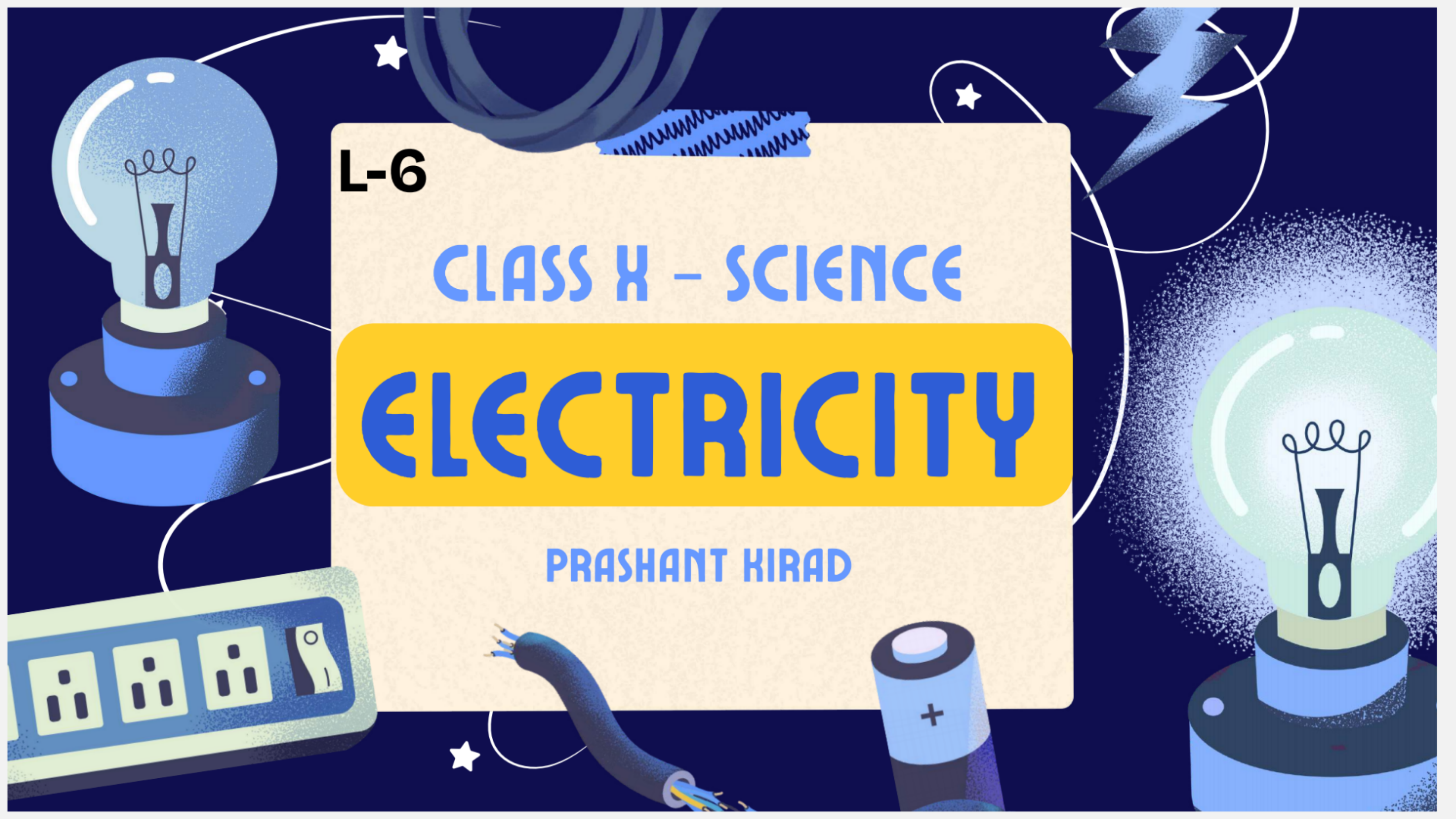




L-6

CLASS X – SCIENCE

ELECTRICITY

PRASHANT KIRAD

Q. (a) How does use of a fuse wire protect electrical appliances?

(b) Why does the cord of an electric oven not glow while its heating element does?

(c) Why are alloys commonly used in electrical heating devices? Give reason.

(a) A fuse wire is connected in series with the electrical circuit. When excessive current flows due to overloading or short-circuiting, the fuse wire becomes hot and melts because of its low melting point. This breaks the circuit and protects the appliance from damage.

(b) The heating element of an electric oven has a much higher resistance than its connecting cord. Therefore, it produces more heat due to the heating effect of current ($H = I^2Rt$) and glows. The cord has very low resistance, so it produces negligible heat and does not glow.

(c) Alloys are commonly used in electrical heating devices because they:

- have high resistivity,
- have a high melting point, and
- do not oxidise or burn easily at high temperatures.



Q. Which of these materials would you add as the device in the heater elements of an electric heating device?

(A) Aluminium

☒ (B) Nichrome

(C) Copper

(D) Silver

Q. An electric fuse protects electrical appliances by:

A. Increasing current flow

B. Decreasing voltage

☒ **C. Melting due to excessive heat produced by overcurrent**

D. Storing electrical energy

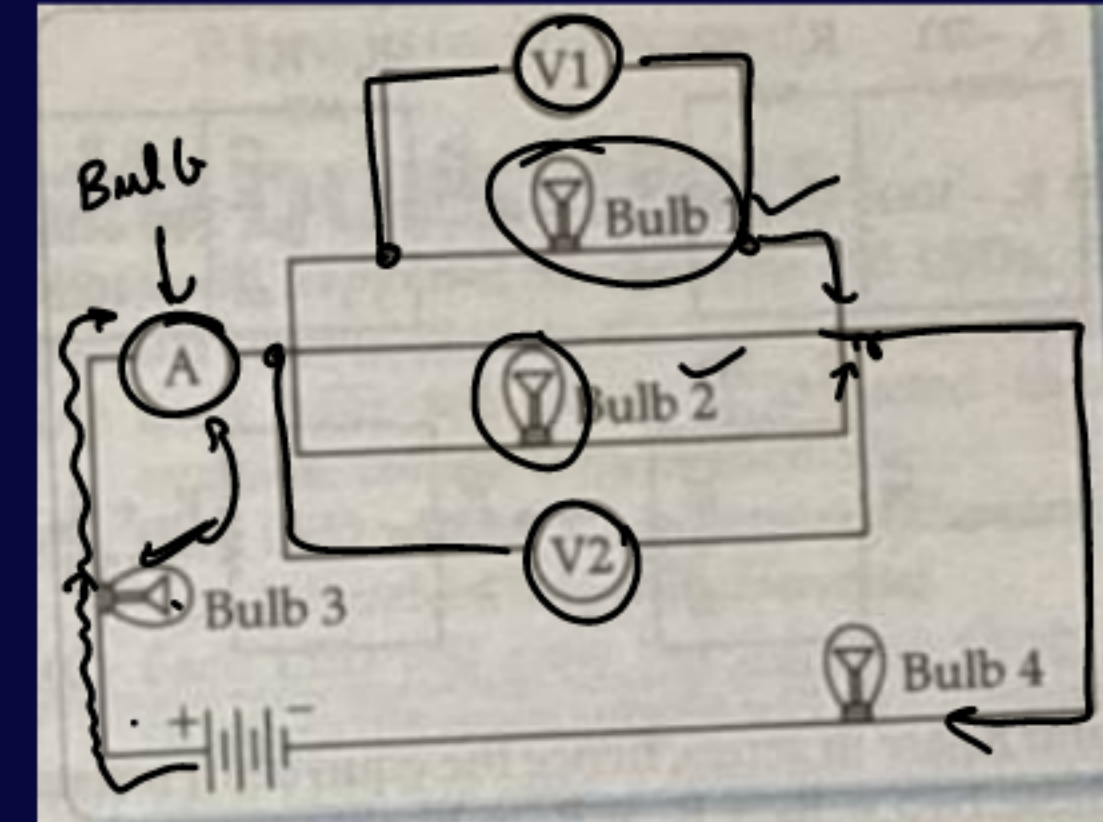
Q. The heating element of an electric heater should have:

- A. Low resistance and low melting point**
- B. High resistance and high melting point**
- C. Low resistance and high conductivity**
- D. High conductivity and low melting point**

Q. Answer the questions based on the electric circuit shown below.
All the four bulbs are identical.

(a) How does the voltage reading on voltmeter V1 compare with the voltage reading on voltmeter V2?

(b) Identify the bulb(s) through which a current equal to the reading on the ammeter flows.

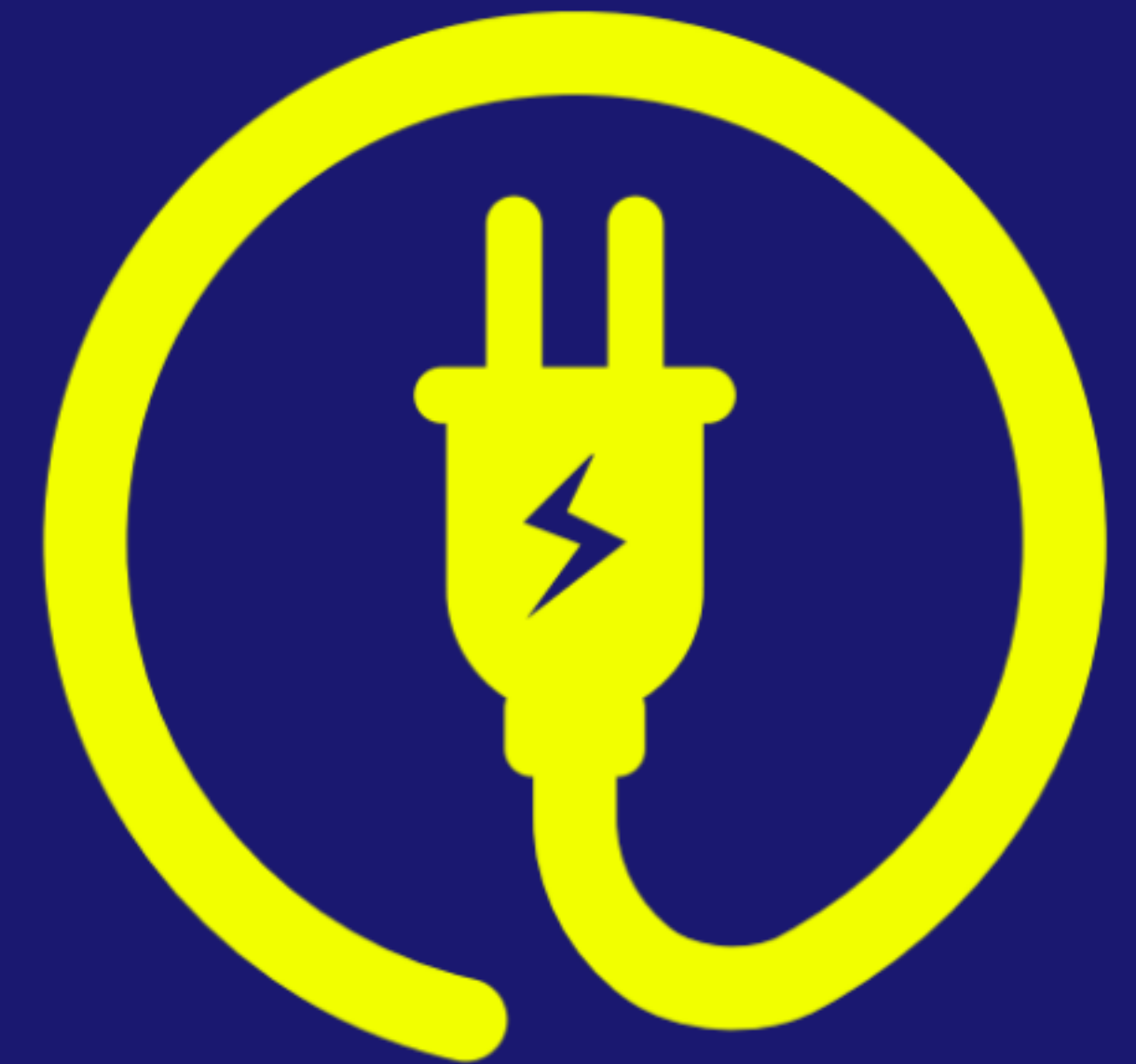


ELECTRIC POWER

✓ Electric power is the rate at which electrical energy is consumed or dissipated in an electric circuit.

In simple words, it tells us how fast an electrical device uses energy.

$$P = VI$$
$$P = I^2 R = V^2 / R$$



$$\text{Power} = \frac{W}{t} = \frac{E}{t}$$

$$P = \frac{W}{t} =$$

$$[W = V \times Q]$$

$$\frac{V \times Q}{t}$$

$$= \boxed{V I}$$

$$V = IR \quad \boxed{P = I^2 R}$$

$$I = \frac{V}{R} \quad \boxed{P = \frac{V^2}{R}}$$

→ Watt

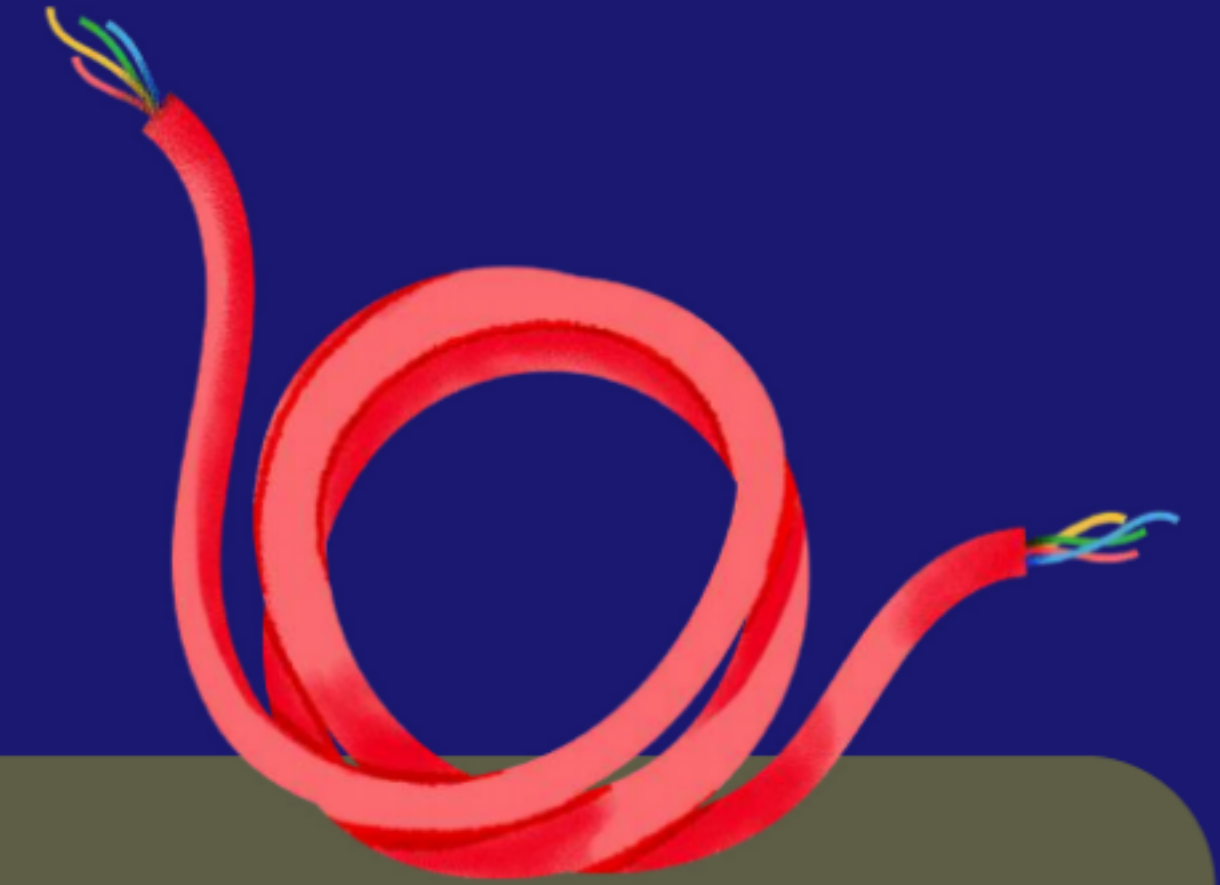
The SI unit of electric power is watt, written as W.

Power consumed by a device is 1 watt if it carries 1 A current when operated at a potential difference of 1 V.)

$$1W = 1V \times 1A$$

$$1W = 1VA$$

Watt



Watt is a small unit, so for larger power we use kilowatt (kW).

$$1kW = 1000W$$

$$P = VI$$

↓

$$1 \text{ watt} = \underline{1V \times 1A}$$

NCERT

Q. An electric bulb is connected to a 220 V generator. The current is 0.50 A. What is the power of the bulb?

$$V = 220 \text{ V}$$

$$I = 0.5$$

$$P = VI$$

$$= 220 \times 0.5$$

$$= \underline{110 \text{ W}}$$

NCERT

Q. An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be –

(a) 100 W

(b) 75 W

~~(c) 50 W~~

~~(d) 25 W~~

~~$P = VI$~~

Rating
↓
Voltage
↓
P

$$P = 100 \text{ W}$$

$$P' = \frac{P}{4} = \frac{100}{4} = \underline{25 \text{ W}}$$

$$P = V^2$$

$$\textcircled{R} \rightarrow$$

$$P \propto V^2$$

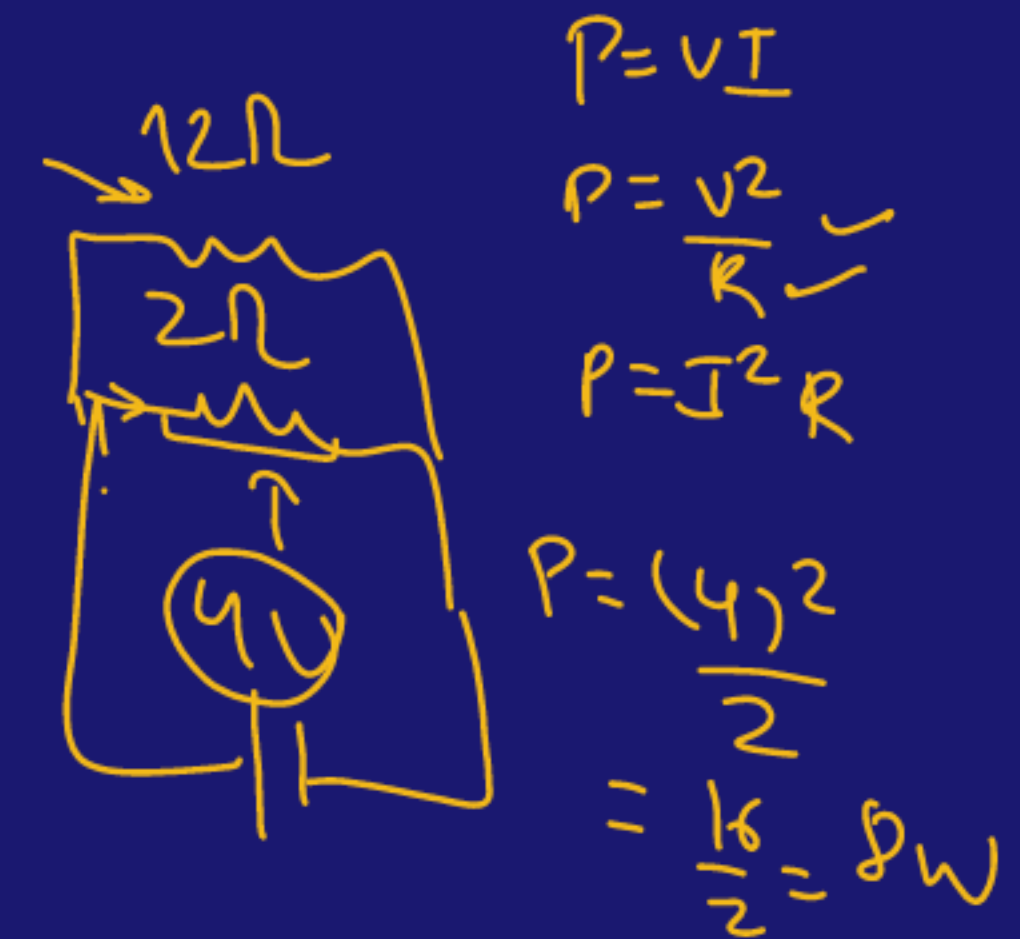
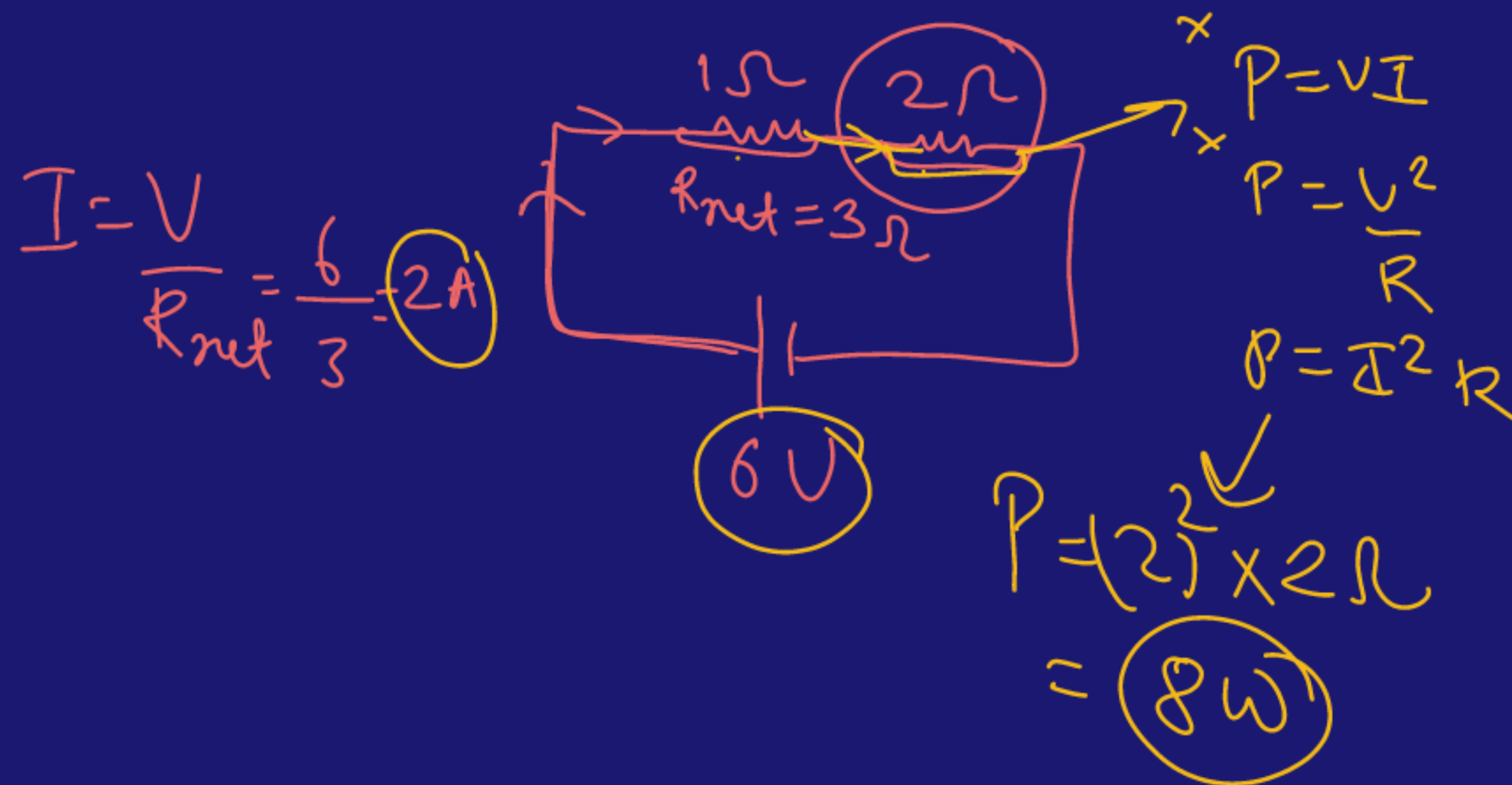
$$\left(\frac{V}{2}\right)^2 = \frac{V^2}{4}$$

NCERT

Q. Compare the power used in the 2 Ω resistor in each of the following circuits:

(i) a 6 V battery in series with 1 Ω and 2 Ω resistors, and

(ii) a 4 V battery in parallel with 12 Ω and 2 Ω resistors.

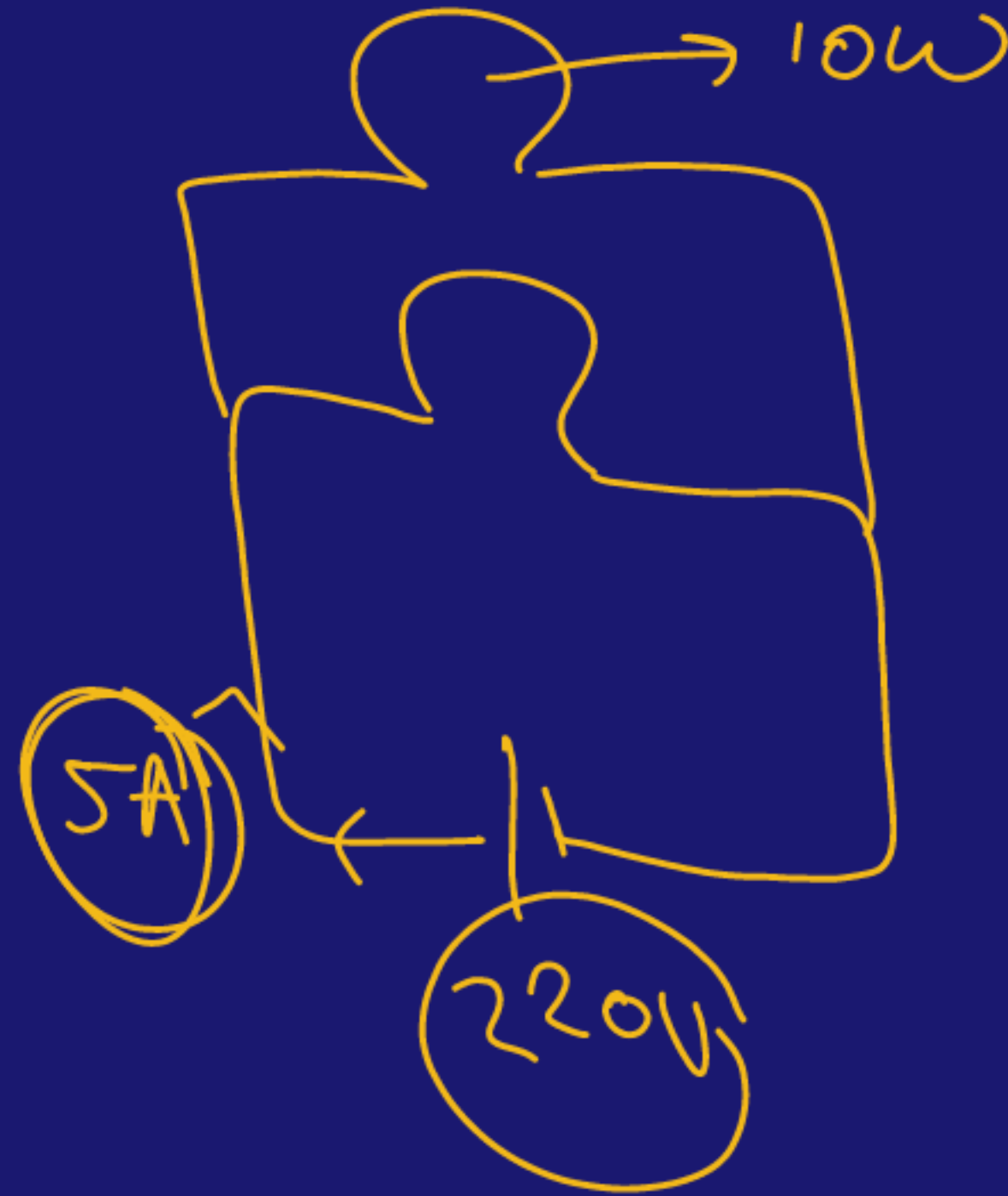


NCERT

Q. Several electric bulbs designed to be used on a 220 V electric supply line, are rated 10 W. How many lamps can be connected in parallel with each other across the two wires of 220 V line if the maximum allowable current is 5 A?

$$\frac{1}{22} \times n = 5 \text{ A}$$

$$n = 5 \times 22 \\ = 110$$



$$P = VI$$

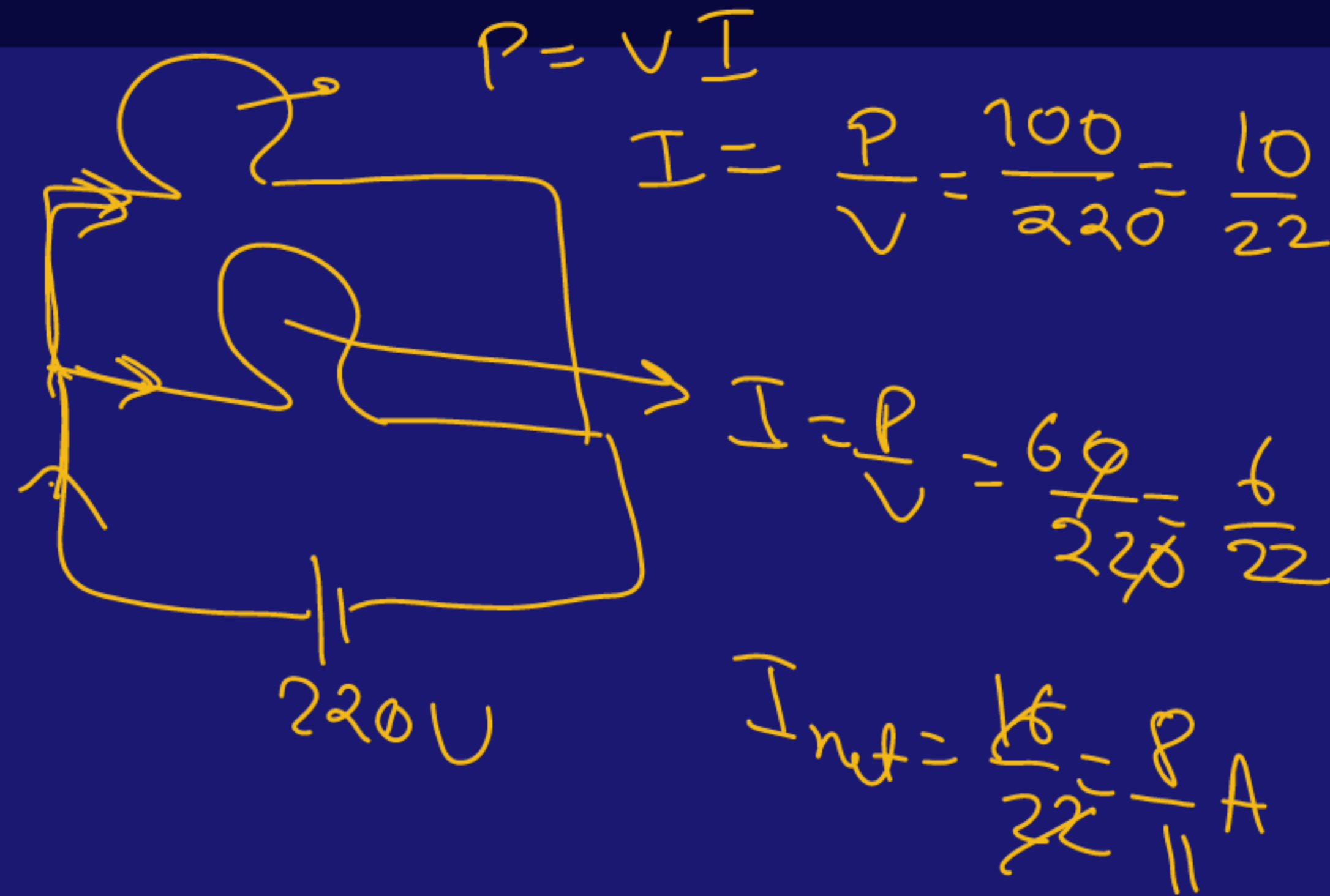
$$10 \text{ W} = 220 \times I$$

$$\frac{10}{220} = I$$

$$I = \frac{1}{22} \text{ A}$$

NCERT

Q. Two lamps, one rated 100 W at 220 V, and the other 60 W at 220 V, are connected in parallel to electric mains supply. What current is drawn from the line if the supply voltage is 220 V?



NCERT

Q. An electric heater of resistance $44\ \Omega$ draws 5 A from the service mains for 2 hours.
Calculate the rate at which heat is developed in the heater.

HW

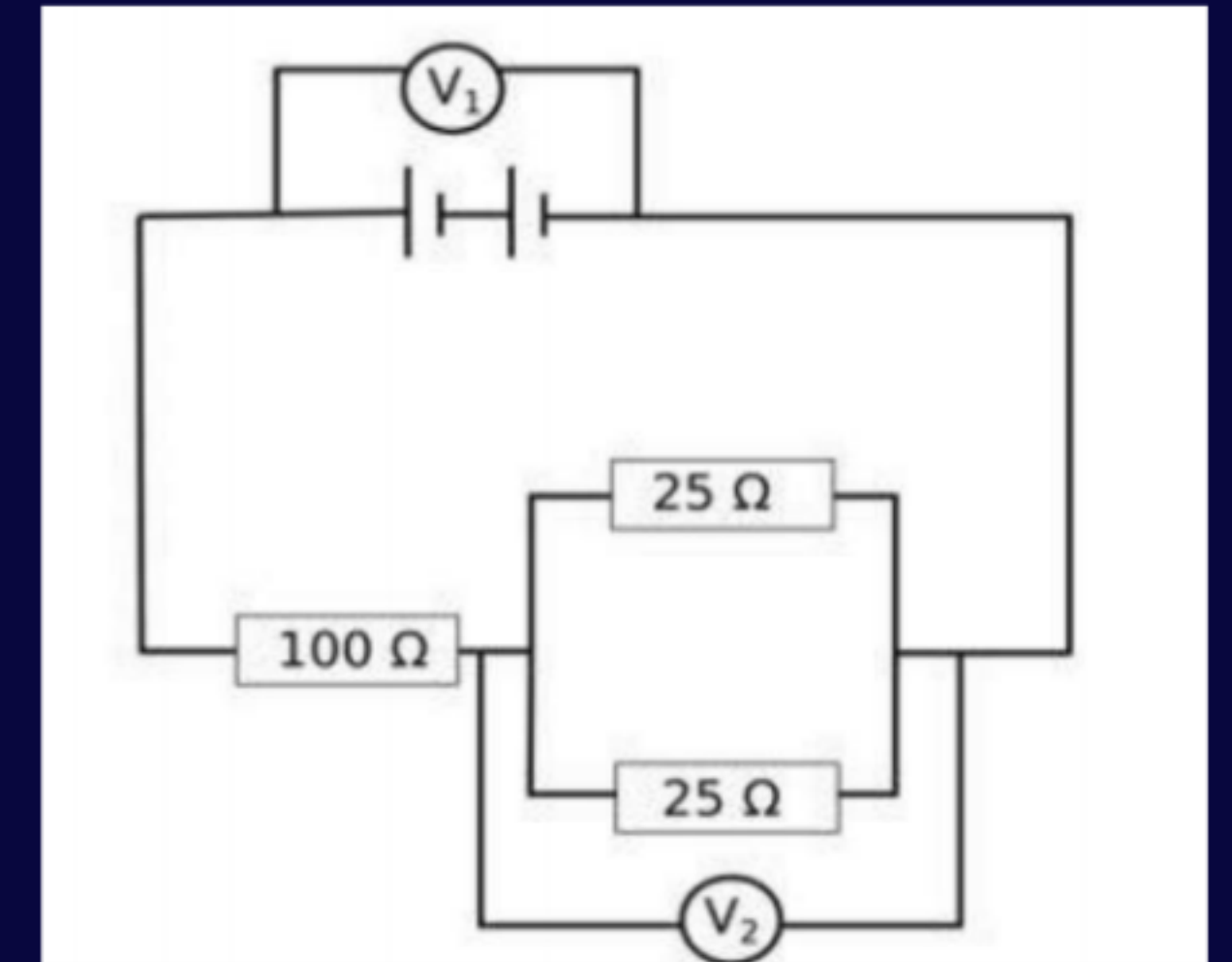
[NCERT]

Q. An electric iron consumes energy at a rate of 840 W when heating is at the maximum rate and 360 W when the heating is at the minimum. The voltage is 220 V. What are the current and the resistance in each case?

Hw

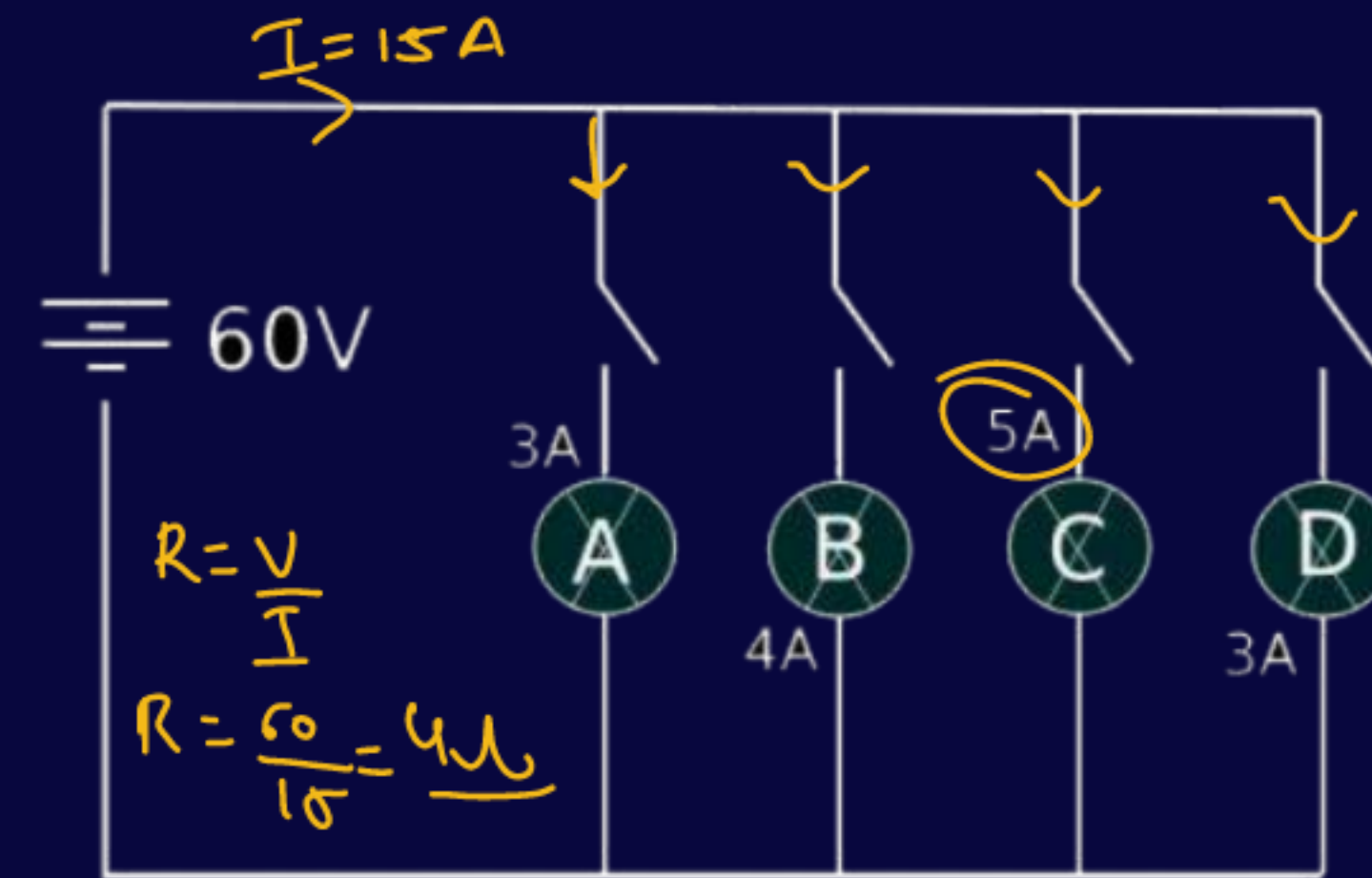
- Q. The arrangement of resistors shown in the above figure is connected to a battery.; The power dissipation in the $100\ \Omega$ resistor is 81 W . Calculate
- (i) the current in the circuit
 - (ii) the reading in the voltmeter V_2
 - (iii) the reading in the voltmeter V_1

kw



Q. In the given circuit, A, B, C and D are four lamps connected with a battery of 60 V. Analyse the circuit and answer the following questions:

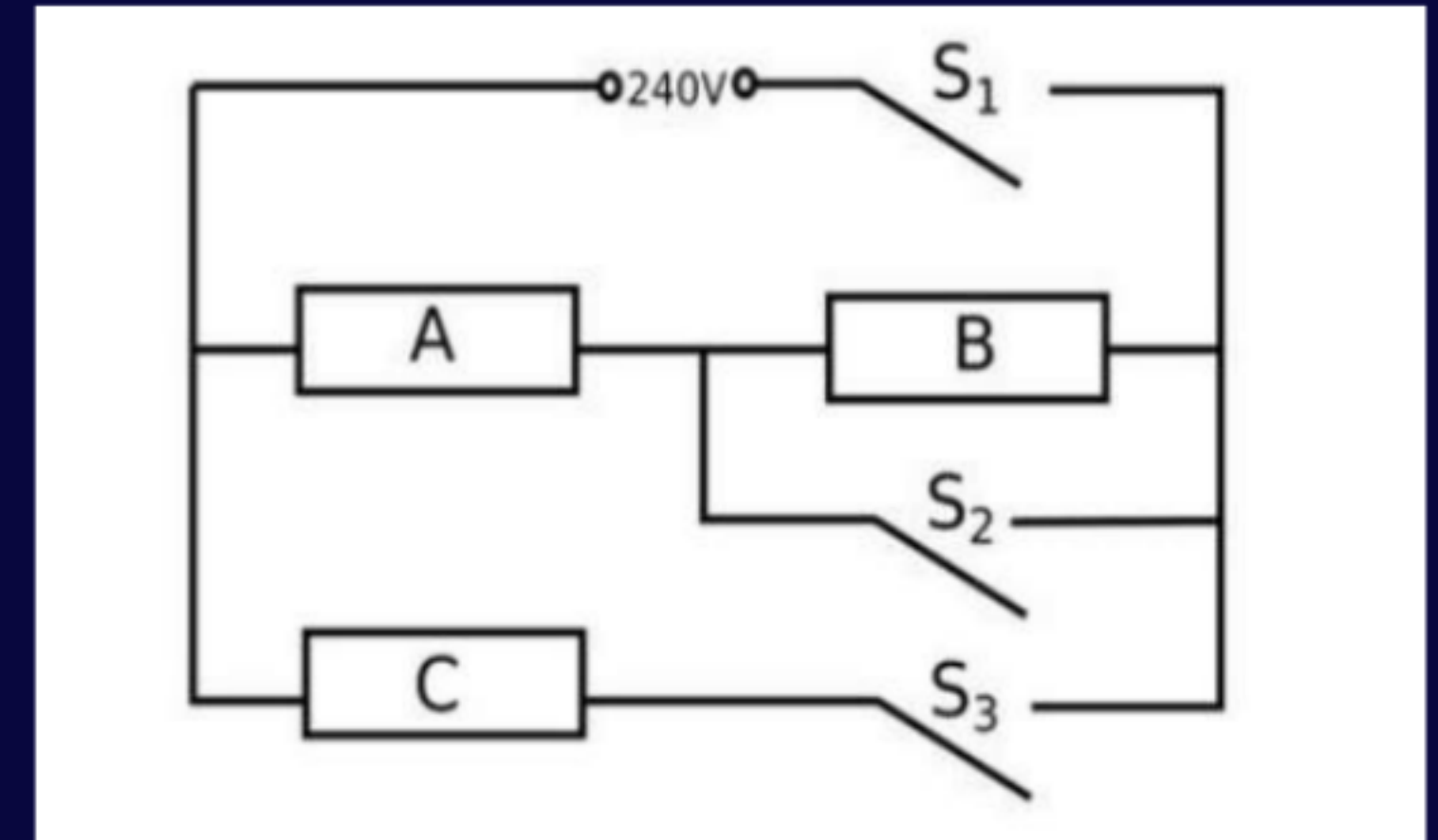
- (i) What kind of combination are the lamps arranged in — series or parallel?
- (ii) Explain with reference to your above answer, any two advantages of this combination of lamps.
- (iii) Explain with proper calculations which lamp glows the brightest.
- (iv) Find out the total resistance of the circuit.



$P \rightarrow \text{Highest}$
 $P = V \times I$

Q. An electric heater consists of three similar heating elements A, B and C, connected as shown in the figure above. Each heating element is rated as 1.2 kW, 240 V and has constant resistance. S1, S2 and S3 are respective switches. The circuit is connected to a 240 V supply.

- (i) Calculate the resistance of one heating element.
- (ii) Calculate the current in each resistor when only S1 and S3 are closed.
- (iii) Calculate the power dissipated across A when S1, S2 and S3 are closed.



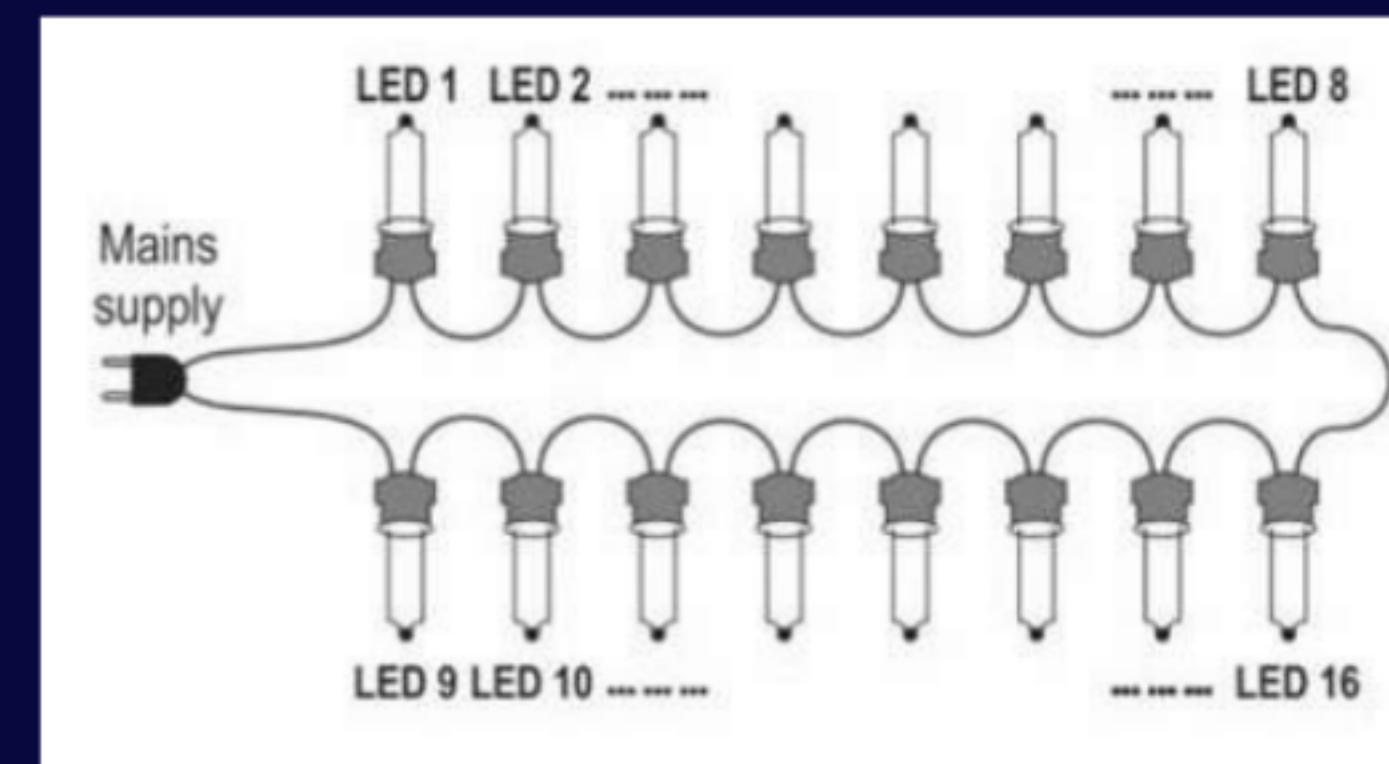
Q.

Read the information given below and answer following questions.

Suresh bought a packet of 100 LEDs to make his own lights for decoration in his house. The packet on the LEDs had the following printed on a label:

LED 2835, 0.2 W, 30 Lumens, 3 V

To understand how he should connect the LEDs, he referred to the following circuit diagram on a website.



Q1. Which of the following describes how the LEDs are connected in the circuit diagram?

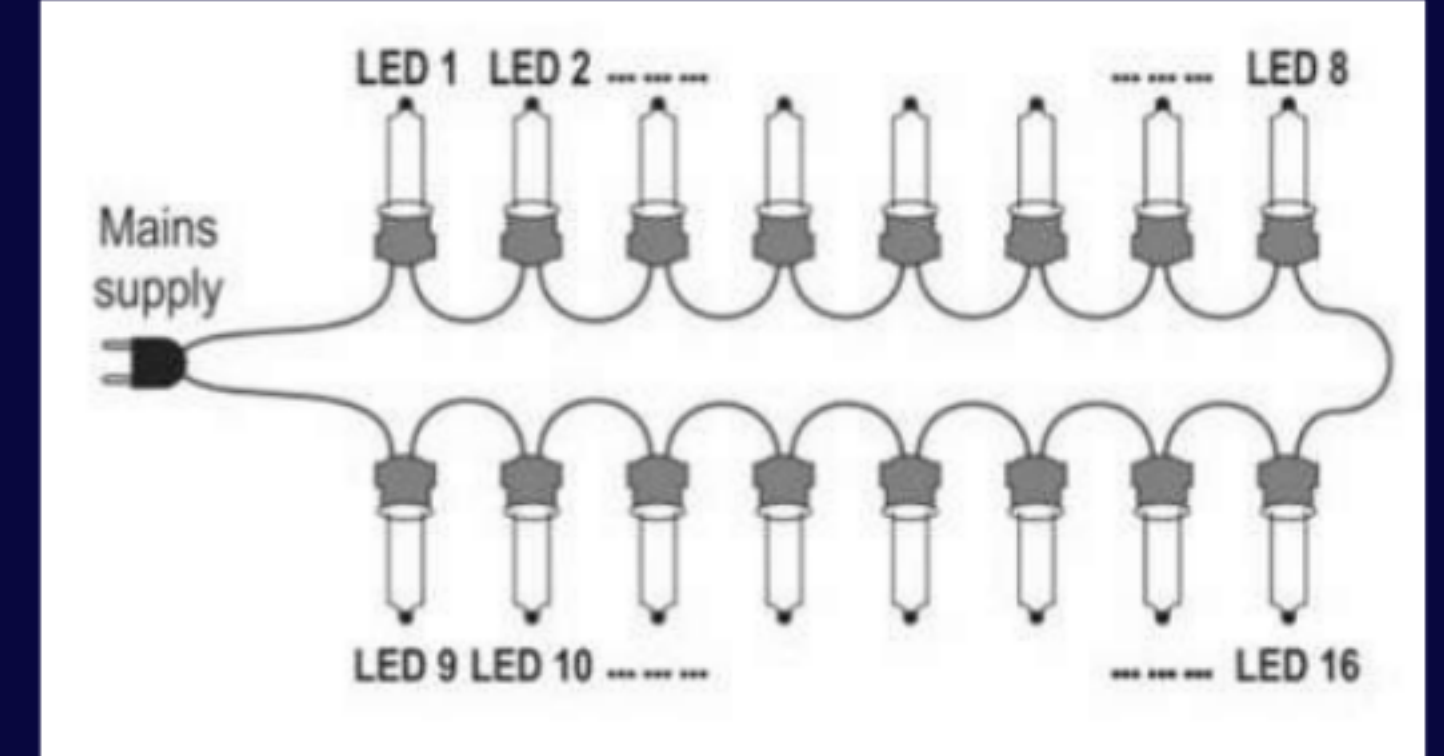
- (A) All in series
- (B) All in parallel
- (C) 8 each in a series combination, and the two combinations in parallel
- (D) 8 each in a parallel combination, and the two combinations in series

Q2. If the LED marked 'LED 2' in the diagram stops working, which other LEDs will also stop working?

- (A) Only LED 3 to LED 8
- (B) Only LED 3 to LED 8 and LED 1
- (C) All the other LEDs in the circuit
- (D) None of the other LEDs in the circuit

Q3. Suresh decided to connect all the LEDs in his lights in a series combination. How many LEDs will he need to connect if he is going to connect the lights to a 240 V mains supply so that the LEDs work at their power rating?

- (A) 16**
- (B) 80**
- (C) 240**
- (D) 1200**



Q4. What will happen if he connects 100 LEDs, all in a parallel combination, to the 240 V mains supply?

- (A) Each LED will work as expected since the available voltage is more than 3 V.**
- (B) Each LED will have a potential difference of 220 V and therefore they will get damaged.**
- (C) Each LED will glow but the ones closer in the circuit to the main supply will glow brighter.**
- (D) Each LED will have a potential difference of 2.4 V across it and therefore will glow dimmer than normal.**